

Spring REST - Hands On

HTTP Request Response

Sample request captured from RFC 7230

```
GET /hello.txt HTTP/1.1
User-Agent: curl/7.16.3 libcurl/7.16.3 OpenSSL/0.9.7l zlib/1.2.3
Host: www.example.com
Accept-Language: en, mi
```

Sample response captured from RFC 7230

```
HTTP/1.1 200 OK
Date: Mon, 27 Jul 2009 12:28:53 GMT
Server: Apache
Last-Modified: Wed, 22 Jul 2009 19:15:56 GMT
ETag: "34aa387-d-1568eb00"
Accept-Ranges: bytes
Content-Length: 51
Vary: Accept-Encoding
Content-Type: text/plain
```

Hello World! My payload includes a trailing CRLF.

Output - Chrome DevTools Network tab, Request Headers on google.com

```
Request URL: https://www.google.com/
Request Method: GET
Status Code: 200 OK
Remote Address: 142.250.193.68:443
Referrer Policy: strict-origin-when-cross-origin

User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36
Accept: text/html,application/xhtml+xml,application/xml;q=0.9
Accept-Language: en-IN,en;q=0.9
```

Hello World RESTful Web Service

HelloController.java

```
package com.cognizant.springlearn.controller;

import org.slf4j.Logger;
import org.slf4j.LoggerFactory;
import org.springframework.web.bind.annotation.GetMapping;
import org.springframework.web.bind.annotation.RestController;

@RestController
public class HelloController {

    private static final Logger LOGGER = LoggerFactory.getLogger(HelloController.class);

    @GetMapping("/hello")
    public String sayHello() {
        LOGGER.info("START");
        String message = "Hello World!!";
        LOGGER.info("END");
        return message;
    }
}
```

application.properties

```
server.port=8083
```

Output - browser, http://localhost:8083/hello

```
Hello World!!
```

Output - Postman GET http://localhost:8083/hello

```
Status: 200 OK
Body:
Hello World!!

Headers:
Content-Type: text/plain;charset=UTF-8
Content-Length: 13
Date: Tue, 21 Jul 2026 06:41:12 GMT
```

REST - Country Web Service

CountryController.java

```
package com.cognizant.springlearn.controller;

import org.slf4j.Logger;
import org.slf4j.LoggerFactory;
import org.springframework.context.ApplicationContext;
import org.springframework.context.support.ClassPathXmlApplicationContext;
import org.springframework.web.bind.annotation.RequestMapping;
import org.springframework.web.bind.annotation.RestController;
import com.cognizant.springlearn.Country;

@RestController
public class CountryController {

    private static final Logger LOGGER = LoggerFactory.getLogger(CountryController.class);

    public CountryController() {
        LOGGER.debug("Inside CountryController Constructor");
    }

    @RequestMapping("/country")
    public Country getCountryIndia() {
        LOGGER.info("START");
        ApplicationContext context = new ClassPathXmlApplicationContext("country.xml");
        Country country = context.getBean("in", Country.class);
        LOGGER.info("END");
        return country;
    }
}
```

Output - browser, http://localhost:8083/country

```
{
  "code": "IN",
  "name": "India"
}
```

Output - Postman Headers tab

```
Content-Type: application/json
Transfer-Encoding: chunked
Date: Tue, 21 Jul 2026 06:44:31 GMT
Keep-Alive: timeout=60
```

Connection: keep-alive

REST - Get all countries

CountryController.java - getAllCountries() added

```
@SuppressWarnings("unchecked")
@GetMapping("/countries")
public List<Country> getAllCountries() {
    LOGGER.info("START");
    ApplicationContext context = new ClassPathXmlApplicationContext("country.xml");
    List<Country> countryList = (List<Country>) context.getBean("countryList");
    LOGGER.info("END");
    return countryList;
}
```

Output - GET http://localhost:8083/countries

```
[
  { "code": "IN", "name": "India" },
  { "code": "US", "name": "United States" },
  { "code": "DE", "name": "Germany" },
  { "code": "JP", "name": "Japan" }
]
```

REST - Get country based on country code

CountryService.java

```
package com.cognizant.springlearn.service;

import java.util.List;
import org.slf4j.Logger;
import org.slf4j.LoggerFactory;
import org.springframework.context.ApplicationContext;
import org.springframework.context.support.ClassPathXmlApplicationContext;
import org.springframework.stereotype.Service;
import com.cognizant.springlearn.Country;

@Service
public class CountryService {

    private static final Logger LOGGER = LoggerFactory.getLogger(CountryService.class);

    @SuppressWarnings("unchecked")
    public Country getCountry(String code) {
        LOGGER.info("START");
        ApplicationContext context = new ClassPathXmlApplicationContext("country.xml");
        List<Country> countryList = (List<Country>) context.getBean("countryList");

        Country result = countryList.stream()
            .filter(c -> c.getCode().equalsIgnoreCase(code))
            .findFirst()
            .orElse(null);

        LOGGER.info("END");
        return result;
    }
}
```

CountryController.java - getCountry() added

```

@Autowired
private CountryService countryService;

@GetMapping("/countries/{code}")
public Country getCountry(@PathVariable String code) {
    LOGGER.info("START");
    Country country = countryService.getCountry(code);
    LOGGER.info("END");
    return country;
}

```

Output - GET http://localhost:8083/countries/in

```

{
  "code": "IN",
  "name": "India"
}

```

REST - Get country exceptional scenario

CountryNotFoundException.java

```

package com.cognizant.springlearn.service.exception;

import org.springframework.http.HttpStatus;
import org.springframework.web.bind.annotation.ResponseStatus;

@ResponseStatus(value = HttpStatus.NOT_FOUND, reason = "Country not found")
public class CountryNotFoundException extends RuntimeException {

    private static final long serialVersionUID = 1L;
}

```

CountryService.java - getCountry() updated to throw exception

```

public Country getCountry(String code) {
    LOGGER.info("START");
    ApplicationContext context = new ClassPathXmlApplicationContext("country.xml");
    List<Country> countryList = (List<Country>) context.getBean("countryList");

    Country result = countryList.stream()
        .filter(c -> c.getCode().equalsIgnoreCase(code))
        .findFirst()
        .orElseThrow(CountryNotFoundException::new);

    LOGGER.info("END");
    return result;
}

```

CountryController.java - throws clause added

```

@GetMapping("/countries/{code}")
public Country getCountry(@PathVariable String code) throws CountryNotFoundException {
    LOGGER.info("START");
    Country country = countryService.getCountry(code);
    LOGGER.info("END");
    return country;
}

```

Output - curl -i http://localhost:8083/country/az

```

HTTP/1.1 404
Content-Type: application/json
Transfer-Encoding: chunked

```

Date: Tue, 21 Jul 2026 06:52:07 GMT

```
{
  "timestamp": "2026-07-21T06:52:07.521+0000",
  "status": 404,
  "error": "Not Found",
  "message": "Country not found",
  "path": "/country/az"
}
```

MockMVC - Test get country service

SpringLearnApplicationTests.java

```
package com.cognizant.springlearn;

import static org.junit.Assert.assertNotNull;
import static org.springframework.test.web.servlet.request.MockMvcRequestBuilders.get;
import static org.springframework.test.web.servlet.result.MockMvcResultMatchers.status;
import static org.springframework.test.web.servlet.result.MockMvcResultMatchers.jsonPath;

import org.junit.Test;
import org.junit.runner.RunWith;
import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.boot.test.autoconfigure.web.servlet.AutoConfigureMockMvc;
import org.springframework.boot.test.context.SpringBootTest;
import org.springframework.test.context.junit4.SpringRunner;
import org.springframework.test.web.servlet.MockMvc;
import org.springframework.test.web.servlet.ResultActions;

import com.cognizant.springlearn.controller.CountryController;

@RunWith(SpringRunner.class)
@SpringBootTest
@AutoConfigureMockMvc
public class SpringLearnApplicationTests {

    @Autowired
    private CountryController countryController;

    @Autowired
    private MockMvc mvc;

    @Test
    public void contextLoads() {
        assertNotNull(countryController);
    }

    @Test
    public void testGetCountry() throws Exception {
        ResultActions actions = mvc.perform(get("/country"));
        actions.andExpect(status().isOk());
        actions.andExpect(jsonPath("$.code").exists());
        actions.andExpect(jsonPath("$.code").value("IN"));
        actions.andExpect(jsonPath("$.name").exists());
        actions.andExpect(jsonPath("$.name").value("India"));
    }
}
```

Output - run as JUnit Test in Eclipse

```

240715|12:08:41.552|main          |DEBUG|CountryController  |
<init>|Inside CountryController Constructor
240715|12:08:42.017|main          |INFO |CountryController  |
getCountryIndia|START
240715|12:08:42.061|main          |INFO |CountryController  |
getCountryIndia|END

JUnit
Tests run: 2, Failures: 0, Errors: 0, Skipped: 0

```

Output - mvn clean test

```

[INFO] -----
[INFO]  T E S T S
[INFO] -----
[INFO] Running com.cognizant.springlearn.SpringLearnApplicationTests
[INFO] Tests run: 2, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 4.117 s
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 2, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] BUILD SUCCESS

```

MockMVC - Test get country service for exceptional scenario

SpringLearnApplicationTests.java - testGetCountryException() added

```

@Test
public void testGetCountryException() throws Exception {
    ResultActions actions = mvc.perform(get("/countries/az"));
    actions.andExpect(status().isNotFound());
    actions.andExpect(status().reason("Country not found"));
}

```

Output - run as JUnit Test in Eclipse

```

240715|12:15:26.304|main          |INFO |CountryService     |
getCountry|START
240715|12:15:26.309|main          |INFO |CountryService     |
getCountry|END

JUnit
Tests run: 3, Failures: 0, Errors: 0, Skipped: 0

```